

# Yan Xie

[yanxieyx@umich.edu](mailto:yanxieyx@umich.edu) | <https://yxie1010.github.io/yan-website/>

## ◆ Education

**University of Michigan, Ann Arbor**

Aug 2019 – May 2025

Ph.D., Climate and Space Sciences and Engineering (GPA: 4.0 / 4.0)

*A+ in Advanced Fluid Dynamics, Remote Sensing etc.; A's in Machine Learning, Bayesian Data Analysis etc.*

**McGill University**

Jul – Oct 2018

Undergraduate Research Trainee

**Nanjing University**

Sep 2015 – June 2019

Bachelor's Degree in Atmospheric Science, Graduate with honors (GPA: 3.8 / 4.0)

## ◆ Employment

**University of Oklahoma, Norman**

Jul 2025 - present

Postdoctoral Research Associate

**University of Michigan, Ann Arbor**

May – June 2025

Research Fellow

## ◆ Service & Mentorship

**Panelist** for NASA's 2025 ROSES Earth Venture research proposal review panel

Spring 2026

**Reviewer** for journals including *JGR-Atmospheres*, *Artificial Intelligence for the Earth Systems*, *Journal of*

*Hydrology*, *Water Resources Research*, *Atmospheric Chemistry and Physics*, etc.

since Mar 2025

**Graduate Student Peer Mentor** – Department of Climate and Space Sciences and Engineering

Fall 2023

**Graduate Student Instructor** – Department of Climate and Space Sciences and Engineering

Winter 2023

## ◆ Honor and Membership

American Geophysical Union Outstanding Student Presentation Award (top 3-5%)

March 2025

Rackham Predoctoral Fellowship – University of Michigan

March 2024

American Geophysical Union – Precipitation Technical Committee

since January 2024

Michigan Geophysical Union – Organization Committee

January– April 2023

## ◆ Publications

King, F., Pettersen, C., Posselt, D., Ringerud, S., & Xie, Y. (2025). Leveraging Sparse Autoencoders to Reveal Interpretable Features in Geophysical Models. *Journal of Geophysical Research: Machine Learning and Computation*, doi: [10.1029/2025JH000769](https://doi.org/10.1029/2025JH000769)

Xie, Y., King, F., Pettersen, C., & Flanner, M. (2025). Machine Learning detection of melting layers from radar observations. *Journal of Geophysical Research: Machine Learning and Computation*, doi:

[10.1029/2024JH000521](https://doi.org/10.1029/2024JH000521)

Xie, Y., Pettersen, C., Flanner, M., & Shates, J. (2024). Ground-observed snow albedo changes during rain-on-snow events in northern Alaska. *Journal of Geophysical Research: Atmospheres*, 129, doi:

[10.1029/2024JD040975](https://doi.org/10.1029/2024JD040975)

Liu, Y., Huang, Y., Yuan, J., Xie, Y., & Zhou, C. (2024). Contribution of surface radiative effects, heat fluxes and their interactions to land surface temperature variability. *Journal of Geophysical Research:*

*Atmospheres*, 129, doi:[10.1029/2023JD039495](https://doi.org/10.1029/2023JD039495)

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., & Drouin, B. J. (2023). Joint use of far-infrared and mid-infrared observation for sounding retrievals: Learning from the past for upcoming far-infrared missions. *Earth and Space Science*, 10, doi:[10.1029/2022EA002684](https://doi.org/10.1029/2022EA002684)

**Xie, Y.,** Huang, X., Chen, X., L’Ecuyer, T. S., Drouin, B. J., & Wang, J. (2022). Retrieval of Surface Spectral Emissivity in Polar Regions Based on the Optimal Estimation Method. *Journal of Geophysical Research: Atmospheres*, 127, doi:[10.1029/2021JD035677](https://doi.org/10.1029/2021JD035677)

L’Ecuyer, T. S., Drouin, B. J., Anheuser, J., Grames, M., Henderson, D. S., Huang, X., Kahn, B. H., Kay, J. E., Lim, B. H., Mateling, M., Merrelli, A., Miller, N. B., Padmanabhan, S., Peterson, C., Schlegel, N., White, M. L., & **Xie, Y.** (2021). The Polar Radiant Energy in the Far Infrared Experiment: A New Perspective on Polar Longwave Energy Exchanges, *Bulletin of the American Meteorological Society*, 102(7), doi:[10.1175/bams-d-20-0155.1](https://doi.org/10.1175/bams-d-20-0155.1)

Huang, Y., Chou, G., **Xie, Y.**, & Soulard, N. (2019). Radiative Control of the Interannual Variability of Arctic Sea Ice. *Geophysical Research Letters*, 46, 9899–9908, doi: [10.1029/2019gl084204](https://doi.org/10.1029/2019gl084204)

#### ◆ Presentations

**Xie, Y.,** McGovern, A., Bostrom, A., & Rao, D. “Bias correction in data preparation and its implications for AI models”. **American Meteorological Society 106<sup>th</sup> Annual Meeting**. 2026, Texas (Oral Presentation).

**Xie, Y.,** King, F., Pettersen, C., & Flanner, M. “Melting Layer Detection from Radar Observations Using Machine Learning”. **American Geophysical Union 2024 Fall Meeting**. (H34H-02 Oral Presentation **OSPA**)

**Xie, Y.,** Pettersen, C., Flanner, M., & Shates, J. “Ground-observed Influence of Rainfall on Surface Snow Albedo at North Slope of Alaska”. **American Geophysical Union 2023 Fall Meeting**. (C32C-08 eLightning Presentation)

**Xie, Y.,** Huang, X., Chen, X., L’Ecuyer, T. S., Drouin, B. J. “On the use of far-IR radiances in satellite retrievals: how can the observations collected half century ago help us preparing for the upcoming missions”. **American Geophysical Union 2022 Fall Meeting**. (A32B-03 Oral Presentation)

**Xie, Y.,** Huang, X. and Chen., X. “Retrieval of surface spectral emissivity in the polar regions: an optimal-estimation approach”. **American Geophysical Union 2020 Fall Meeting**. (A239-07 Oral Presentation)

#### ◆ Research Experiences

##### **Bias correction of surface temperature extremes for improved AI forecasts of heatwaves**

Jul 2025 – Present Postdoctoral Research Associate School of Meteorology, University of Oklahoma  
Mentors: Prof. Amy McGovern, Prof. Ann Bostrom

- Developed methods to identify and correct data biases in AI weather models’ training data
- Assessed the influence of bias correction on AI-based forecasts of high-impact heat wave events

##### **Investigation melting layers from ground-based radar observations leveraging machine learning**

Jan 2024 – Jun 2025 Rackham Predoctoral Fellow Dept. of CLaSP, University of Michigan  
Mentors: Dr. Fraser King, Prof. Claire Pettersen, Prof. Mark Flanner

- Developed a precipitation phase identification methodology to improve detection of impactful rain-on-snow events in northern Alaska
- Evaluated snow hydrology in Community Land Model v5.0 and SNICAR-ADv3
- Developed a novel machine learning model to automatically detect melting layers across diverse weather conditions using radar observations and performed hyperparameter tuning using TensorFlow
- Assessed the machine learning model in terms of its bias, accuracy, robustness, and interpretability

##### **Satellite retrievals of atmospheric profiles and surface properties in polar regions**

Sep 2019 – Sep 2022 Graduate Student Research Assistant Dept. of CLaSP, University of Michigan  
Mentors: Prof. Xianglei Huang, Dr. Xiuhong Chen

- Developed an optimal-estimation (OE) based Bayesian framework to retrieve mid- and far-infrared surface spectral emissivity using satellite observations over Arctic and Antarctic
- Demonstrated the influence of water vapor absorption and spectral resolution on satellite remote sensing

#### ◆ Professional Skills

Python, Tensorflow, PyTorch, MATLAB, R, C, Fortran, NCL, LaTeX